



**COMPETENCY STANDARD**

**FOR**

**ADVANCED PHYSICAL DESIGN PROGRAM:**  
**RTL TO GDSII**

**(SEMICONDUCTOR / VLSI SECTOR)**

**Skills for Industry Competitiveness and Innovation Program (SICIP)**  
**Finance Division, Ministry of Finance**



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The Competency Standards for Advanced Physical Design Program: RTL to GDSII serve as a foundational document for the development of curricula, teaching materials, and assessment tools specifically tailored for the Semiconductor/ VLSI sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant roles within the sector.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP), this document addresses the skills requirements critical for Advanced Physical Design Program: RTL to GDSII. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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## INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

**The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.**

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

## OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various Light Engineering companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.

- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

### Units & Elements at a Glance:

#### Generic Competencies (08 hrs.)

| Code                | Unit of Competency                                    | Elements of Competency                                                                                                                                                                                 | Duration (Hours) |
|---------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| SICIP-SCD-APDP-01-G | Apply OHS, ESD and Safe Workplace Practices           | <ol style="list-style-type: none"> <li>1. Identify OHS and ESD requirements</li> <li>2. Practice safe workstation and EDA lab procedures</li> <li>3. Follow emergency response procedures</li> </ol>   | 04               |
| SICIP-SCD-APDP-02-G | Apply Workplace Communication and Professional Ethics | <ol style="list-style-type: none"> <li>1. Interpret workplace communication and etiquette</li> <li>2. Prepare technical documents</li> <li>3. Participate in design review and presentation</li> </ol> | 04               |
| <b>Total Hour</b>   |                                                       |                                                                                                                                                                                                        | <b>08 Hrs.</b>   |

#### Sector Specific Competencies (32 hrs.)

| Code                | Unit of Competency                                                       | Elements of Competency                                                                                                                                                                                       | Duration (Hours) |
|---------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| SICIP-SCD-APDP-01-S | Interpret Semiconductor Industry, CMOS Fundamentals and ASIC Design Flow | <ol style="list-style-type: none"> <li>1. Describe semiconductor industry and design models</li> <li>2. Apply CMOS and timing fundamentals</li> <li>3. Interpret ASIC design flow and design data</li> </ol> | 16               |
| SICIP-SCD-APDP-02-S | Operate Linux, EDA Tools and Automation Scripts                          | <ol style="list-style-type: none"> <li>1. Operate Linux EDA tools</li> <li>2. Apply shell and TCL scripting</li> <li>3. Use version control systems</li> <li>4. Execute automation scripts</li> </ol>        | 16               |
| <b>Total Hour</b>   |                                                                          |                                                                                                                                                                                                              | <b>32 Hrs.</b>   |

## Occupation Specific Competencies (200 hrs.)

| Code                | Unit of Competency                                     | Elements of Competency                                                                                                                                                                                                                                                                                                                                         | Duration (Hours) |
|---------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| SICIP-SCD-APDP-01-O | Develop Synthesizable RTL using Verilog/System Verilog | <ol style="list-style-type: none"> <li>1. Implement combinational logic.</li> <li>2. Implement sequential and FSM designs.</li> <li>3. Apply RTL coding guidelines.</li> <li>4. Prepare testbench and simulation setup.</li> <li>5. Run simulation and debug waveforms.</li> <li>6. Validate RTL functionality.</li> </ol>                                     | 32               |
| SICIP-SCD-APDP-02-O | Perform Logic Synthesis and Timing Analysis            | <ol style="list-style-type: none"> <li>1. Execute logic synthesis flow.</li> <li>2. Create and debug timing constraints.</li> <li>3. Optimize and analyze QoR.</li> <li>4. Analyze timing fundamentals.</li> <li>5. Perform static timing analysis (STA).</li> <li>6. Apply timing closure techniques.</li> </ol>                                              | 36               |
| SICIP-SCD-APDP-03-O | Carry out Physical Design Implementation               | <ol style="list-style-type: none"> <li>1. Perform floor planning and power planning.</li> <li>2. Perform placement and congestion analysis.</li> <li>3. Perform clock tree synthesis (CTS).</li> <li>4. Perform routing and physical optimization.</li> <li>5. Perform extraction and parasitic analysis.</li> <li>6. Prepare final GDSII database.</li> </ol> | 52               |
| SICIP-SCD-APDP-04-O | Perform Physical Verification and Signoff              | <ol style="list-style-type: none"> <li>1. Perform DRC and LVS verification.</li> <li>2. Perform signoff checks.</li> <li>3. Analyze power components.</li> <li>4. Analyze signal integrity and reliability.</li> <li>5. Review signoff results.</li> </ol>                                                                                                     | 24               |
| SICIP-SCD-APDP-05-O | Execute Advanced Design and Verification Techniques    | <ol style="list-style-type: none"> <li>1. Apply low-power design techniques.</li> <li>2. Apply DFT and ATPG concepts.</li> <li>3. Apply formal verification</li> </ol>                                                                                                                                                                                         | 24               |

|                     |                                              |                                                                                                                                                                                                                                      |               |
|---------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
|                     |                                              | and assertion basics.<br>4. Perform LEC and formal debug.<br>5. Apply advanced flow optimization concepts.<br>6. Interpret HLS fundamentals.                                                                                         |               |
| SICIP-SCD-APDP-06-O | Perform RTL-to-GDSII Physical Design Project | 1. Plan RTL-to-GDSII implementation.<br>2. Execute the complete physical design flow.<br>3. Validate design quality and signoff results.<br>4. Prepare project documentation and deliverables.<br>5. Present and defend design work. | 32            |
| <b>Total Hour</b>   |                                              |                                                                                                                                                                                                                                      | <b>0 Hrs.</b> |



## COMPETENCY STANDARD: ADVANCED PHYSICAL DESIGN PROGRAM: RTL TO GDSII

### The Generic Competencies

|                                                                                                                                                                                                                                                                                                                          |                                         |                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------------------------------|
| <b>Unit of Competency:</b><br><b>APPLY OHS, ESD AND SAFE<br/>WORKPLACE PRACTICES</b>                                                                                                                                                                                                                                     | <b>Nominal<br/>Duration:</b><br>04 hrs. | <b>Unit Code:</b><br>SICIP-SCD-APDP-01-G |
| <b>Unit Descriptor:</b><br><br>This unit covers the knowledge, skills and attitudes required to apply OHS, ESD and Safe Workplace Practices. It specifically includes the tasks of identifying OHS and ESD requirements, practicing safe workstation and EDA lab procedures and following emergency response procedures. |                                         |                                          |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                              | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Identify OHS and ESD requirements                | 1.1 <b><u>Workplace OHS and ESD requirements</u></b> are identified according to organizational procedures.<br>1.2 <b><u>Hazards</u></b> associated with semiconductor and EDA laboratory environments are recognized.<br>1.3 <b><u>Personal protective equipment (PPE) and ESD protection devices</u></b> are identified and used appropriately.<br>1.4 ESD control measures are applied to protect electronic components and equipment.<br>1.5 <b><u>Workplace safety signs, symbols, and instructions</u></b> are interpreted correctly. |
| 2. Practice safe workstation and EDA lab procedures | 2.1 <b><u>Workstations</u></b> are prepared according to OHS and ESD requirements.<br>2.2 <b><u>EDA tools, computers, and laboratory equipment</u></b> are operated following safe work procedures.<br>2.3 <b><u>Cables, electrical equipment, and work areas</u></b> are maintained to prevent workplace hazards.<br>2.4 Good housekeeping practices are maintained to ensure a safe working environment.<br>2.5 Workplace incidents, hazards, and unsafe conditions are identified and reported promptly.                                 |
| 3. Follow emergency response procedures             | 3.1 <b><u>Emergency procedures</u></b> are identified according to workplace requirements.<br>3.2 Emergency exits, assembly points, and safety equipment are located and accessed correctly.<br>3.3 Appropriate actions are taken during electrical, fire, or other workplace emergencies.<br>3.4 Incidents and accidents are reported according to organizational procedures.                                                                                                                                                              |

|  |                                                                                                                   |
|--|-------------------------------------------------------------------------------------------------------------------|
|  | 3.5 <b><u>Emergency response activities</u></b> are carried out in accordance with workplace safety requirements. |
|--|-------------------------------------------------------------------------------------------------------------------|

## Range of Variables

| Variable                                                          | Range<br>May include but not limited to:                                                                                                                      |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Workplace OHS and ESD requirements                             | 1.1 OHS Policies<br>1.2 ESD Procedures<br>1.3 Workplace Rules<br>1.4 Safe Work Practices<br>1.5 Emergency Procedures<br>1.6 Risk Assessment                   |
| 2. Hazards                                                        | 2.1 Electrostatic Discharge (ESD)<br>2.2 Electrical Hazards<br>2.3 Fire Hazards<br>2.4 Chemical Hazards<br>2.5 Trip and Slip Hazards<br>2.6 Equipment Hazards |
| 3. Personal protective equipment (PPE) and ESD protection devices | 3.1 ESD Wrist Strap<br>3.2 ESD Shoes<br>3.3 ESD Mat<br>3.4 Safety Glasses<br>3.5 Protective Gloves<br>3.6 Lab Coat                                            |
| 4. Workplace safety signs, symbols, and instructions              | 4.1 Warning Signs<br>4.2 Mandatory Signs<br>4.3 Prohibition Signs<br>4.4 Emergency Exit Signs<br>4.5 Fire Safety Signs<br>4.6 Hazard Symbols                  |
| 5. Workstations                                                   | 5.1 ESD Workstation<br>5.2 Computer Workstation<br>5.3 Design Workbench<br>5.4 Laboratory Bench<br>5.5 Test Workstation<br>5.6 Simulation Workstation         |
| 6. EDA tools, computers, and laboratory equipment                 | 6.1 EDA Software<br>6.2 Workstation Computer<br>6.3 Server<br>6.4 Oscilloscope<br>6.5 Digital Multimeter<br>6.6 Power Supply                                  |
| 7. Cables, electrical equipment, and work areas                   | 7.1 Power Cables<br>7.2 Network Cables<br>7.3 Extension Cords<br>7.4 Electrical Outlets                                                                       |

|                                  |                                                                                                                                                                    |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | 7.5 Work Benches<br>7.6 Cable Management                                                                                                                           |
| 8. Emergency procedures          | 8.1 Fire Emergency<br>8.2 Electrical Emergency<br>8.3 Medical Emergency<br>8.4 Evacuation Procedure<br>8.5 First Aid Procedure<br>8.6 Incident Reporting Procedure |
| 9. Emergency response activities | 9.1 Evacuation<br>9.2 First Aid<br>9.3 Firefighting<br>9.4 Hazard Isolation<br>9.5 Emergency Notification<br>9.6 Incident Documentation                            |

### Curricular Content Guide:

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Principles of Occupational Health and Safety (OHS) and Electrostatic Discharge (ESD) protection.<br>1.2 Common hazards in semiconductor fabrication, EDA laboratories, and electronic work environments.<br>1.3 Proper use of Personal Protective Equipment (PPE) and ESD protection devices.<br>1.4 ESD control methods and safe handling of electronic components.<br>1.5 Safe operation of computers, EDA tools, electrical equipment, and laboratory workstations.<br>1.6 Workplace housekeeping practices and electrical safety requirements.<br>1.7 Emergency procedures, including fire safety, electrical hazards, evacuation routes.<br>1.8 Interpretation of workplace safety signs, symbols, and organizational safety policies and procedures. |
| 2. Underpinning Skills    | 2.1 Applying workplace OHS and ESD requirements in semiconductor and EDA laboratory environments<br>2.2 Interpreting workplace safety signs, symbols, and safety instructions<br>2.3 Preparing and maintaining safe workstations and EDA laboratory environments<br>2.4 Operating EDA tools, computers, and laboratory equipment safely<br>2.5 Applying good housekeeping practices to maintain a hazard-free workplace<br>2.6 Following emergency response procedures for electrical, fire, and workplace emergencies                                                                                                                                                                                                                                         |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                          |                                                                                                                                                                                                                                                                                        |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | 3.3 Sincerity and honesty to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Good relationships with peers, sub-ordinates and seniors in the workplace            |
| 4. Resource Implications | The following resources must be provided:<br><br>4.1 Workplace (simulated or actual)<br>4.2 Learning/operational manual<br>4.3 PPEs<br>4.4 Firefighting equipment<br>4.5 Emergency response manual<br>4.6 Emergency Systems (Fire alarms, emergency exits, etc.)<br>4.7 First aid kits |

### Assessment Evidence Guide:

|                                   |                                                                                                                                                                                                                                                   |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br><br>1.1 identified OHS and ESD requirements<br>1.2 practiced safe workstation and EDA lab procedures<br>1.3 followed emergency response procedures                                            |
| 2. Methods of Assessment          | Methods of assessment should include:<br><br>2.1 Written test<br>2.2 Practical demonstration<br>2.3 Oral question<br>2.4 Portfolio (optional)                                                                                                     |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training<br>3.2 Assessment should be done by a certified assessor or occupation-specific industry expert. |

|                                                                                                                                                                                                                                                                                                                                     |                                     |                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------|
| <b>Unit of Competency:</b><br><b>APPLY WORKPLACE COMMUNICATION AND PROFESSIONAL ETHICS</b>                                                                                                                                                                                                                                          | <b>Nominal Duration:</b><br>04 hrs. | <b>Unit Code:</b><br>SICIP-SCD-APDP-02-G |
| <b>Unit Descriptor:</b><br><br>This unit covers the knowledge, skills and attitudes required to apply workplace communication and professional ethics. It specifically includes the tasks of interpreting workplace communication and etiquette, preparing technical documents and participating in design review and presentation. |                                     |                                          |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                             | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Interpret workplace communication and etiquette | 1.1 <b><u>Workplace communication methods</u></b> and organizational protocols are identified.<br>1.2 <b><u>Professional etiquette</u></b> is demonstrated during workplace interactions.<br>1.3 Technical information and work instructions are interpreted accurately.<br>1.4 <b><u>Verbal and written communication</u></b> is conducted clearly and professionally.<br>1.5 Team communication and collaboration are maintained according to workplace requirements. |
| 2. Prepare technical documents                     | 2.1 <b><u>Technical information</u></b> is collected and organized according to project requirements.<br>2.2 Design notes, reports, and technical documents are prepared using standard formats.<br>2.3 <b><u>Technical terminology and documentation standards</u></b> are applied correctly.<br>2.4 Documents are reviewed for accuracy, completeness, and consistency.<br>2.5 Final documents are submitted according to organizational procedures.                  |
| 3. Participate in design review and presentation   | 3.1 <b><u>Design data and supporting materials</u></b> are prepared for review meetings.<br>3.2 Design concepts and technical information are presented clearly to stakeholders.<br>3.3 Feedback and review comments are recorded and analyzed.<br>3.4 <b><u>Design modifications</u></b> are incorporated based on review outcomes.<br>3.5 Professional communication and teamwork are demonstrated throughout the review and presentation process.                    |

## Range of Variables

| Variable                                             | Range<br>May Include but not limited to:                                                                                                                 |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Workplace communication methods                   | 1.1 Face-to-Face Communication<br>1.2 Email<br>1.3 Phone Calls<br>1.4 Video Conferencing<br>1.5 Instant Messaging                                        |
| 2. Professional etiquette                            | 2.1 Respectful Behaviour<br>2.2 Professional Language<br>2.3 Punctuality<br>2.4 Active Listening<br>2.5 Appropriate Dress Code<br>2.6 Workplace Ethics   |
| 3. Verbal and written communication                  | 3.1 Meetings<br>3.2 Emails<br>3.3 Reports<br>3.4 Presentations<br>3.5 Memos<br>3.6 Technical Documentation                                               |
| 4. Technical information                             | 4.1 Design Specifications<br>4.2 Project Requirements<br>4.3 Technical Drawings<br>4.4 Test Results<br>4.5 Design Data<br>4.6 Reference Documents        |
| 5. Technical terminology and documentation standards | 5.1 Technical Terms<br>5.2 Industry Standards<br>5.3 Naming Conventions<br>5.4 Document Templates<br>5.5 File Formats<br>5.6 Documentation Guidelines    |
| 6. Design data and supporting materials              | 6.1 Design Drawings<br>6.2 Schematics<br>6.3 Presentation Slides<br>6.4 Technical Reports<br>6.5 Simulation Results<br>6.6 Design Specifications         |
| 7. Design modifications                              | 7.1 Design Updates<br>7.2 Specification Changes<br>7.3 Layout Revisions<br>7.4 Documentation Updates<br>7.5 Error Corrections<br>7.6 Improvement Actions |

**Curricular Content Guide:**

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | <ul style="list-style-type: none"><li>1.1 Principles of workplace communication, professional etiquette, and organizational communication protocols.</li><li>1.2 Methods for interpreting technical information, work instructions, and engineering documentation.</li><li>1.3 Standards and formats for preparing technical reports, design notes, and project documentation.</li><li>1.4 Use of technical terminology, documentation conventions, and version control practices.</li><li>1.5 Principles of effective presentation, technical discussion, and design review processes.</li><li>1.6 Methods for receiving, incorporating technical feedback into design improvements.</li><li>1.7 Teamwork, collaboration, and professional communication practices in engineering.</li></ul> |
| 2. Underpinning Skills    | <ul style="list-style-type: none"><li>2.1 Applying professional workplace communication and organizational etiquette</li><li>2.2 Collaborating with team members to support design and project activities</li><li>2.3 Preparing technical documents, reports, and design notes using standard formats</li><li>2.4 Applying technical terminology and documentation standards correctly</li><li>2.5 Reviewing technical documents for accuracy, completeness, and consistency</li><li>2.6 Presenting design concepts and technical information clearly during review meetings</li><li>2.7 Incorporating feedback to improve design quality and project outcomes</li></ul>                                                                                                                      |
| 3. Underpinning Attitudes | <ul style="list-style-type: none"><li>3.1 Commitment to occupational health and safety</li><li>3.2 Promptness in carrying out activities</li><li>3.3 Sincerity and honesty to duties</li><li>3.4 Environmental concerns</li><li>3.5 Eagerness to learn</li><li>3.6 Tidiness and timeliness</li><li>3.7 Respect for rights of peers and seniors in the workplace</li><li>3.8 Good relationships with peers, sub-ordinates and seniors in the workplace</li></ul>                                                                                                                                                                                                                                                                                                                               |
| 4. Resource Implications  | <p>The following resources must be provided:</p> <ul style="list-style-type: none"><li>4.1 Workplace (simulated or actual)</li><li>4.2 Work books</li><li>4.3 Learning manuals</li><li>4.4 Technical documents</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

### Assessment Evidence Guide:

|                                   |                                                                                                                                                                                                                                                  |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br><br>1.1 interpreted workplace communication and etiquette<br>1.2 prepared technical documents<br>1.3 participated in design review and presentation                                          |
| 2. Methods of Assessment          | Methods of assessment should include:<br><br>2.1 Written test<br>2.2 Practical demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (Optional)                                                                                                 |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training<br>3.2 Assessment should be done by a certified assessor or occupation-specific industry expert |



## Sector Specific Competencies

|                                                                                                                                                                                                                                                                                                                                                               |                                         |                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------------------------------|
| <b>Unit of Competency:</b><br><b>INTERPRET SEMICONDUCTOR<br/>INDUSTRY, CMOS FUNDAMENTALS<br/>AND ASIC DESIGN FLOW</b>                                                                                                                                                                                                                                         | <b>Nominal<br/>Duration:</b><br>16 hrs. | <b>Unit Code:</b><br>SICIP-SCD-APDP-01-S |
| <b>Unit Descriptor:</b><br><br>This unit covers the skills, knowledge and attitudes required to interpret semiconductor industry, CMOS fundamentals and ASIC design flow. It specifically includes the tasks of describing semiconductor industry and design models, applying CMOS and timing fundamentals and interpreting ASIC design flow and design data, |                                         |                                          |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                               | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Describe semiconductor industry and design models | 1.1 <b><u>Semiconductor industry structure and application areas</u></b> are identified.<br>1.2 Semiconductor design, fabrication, packaging, and testing processes are described.<br>1.3 <b><u>Design models</u></b> are distinguished according to application requirements.<br>1.4 Front-end and back-end VLSI design processes are explained.<br>1.5 Semiconductor industry standards, roles, and career pathways are identified. |
| 2. Apply CMOS and timing fundamentals                | 2.1 <b><u>CMOS technology and logic gate operations</u></b> are explained.<br>2.2 <b><u>Basic timing concepts</u></b> and propagation delay are interpreted.<br>2.3 <b><u>Timing parameters</u></b> are analyzed according to digital circuit requirements.<br>2.4 <b><u>CMOS design principles</u></b> are applied to simple digital circuits.<br>2.5 Timing behavior is verified using simulation results and timing reports.       |
| 3. Interpret ASIC design flow and design data        | 3.1 <b><u>ASIC design flow</u></b> is interpreted from RTL to GDSII.<br>3.2 Design files, libraries, and technology data are identified according to project requirements.<br>3.3 Design constraints and timing requirements are interpreted before implementation.<br>3.4 <b><u>EDA tools and design environments</u></b> used in the ASIC flow are identified.<br>3.5 Design reports and output data are reviewed to verify         |

|  |                                    |
|--|------------------------------------|
|  | implementation status and quality. |
|--|------------------------------------|

## Range of Variables

| Variable                                                  | Range<br>May include but not limited to:                                                                                                                      |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Semiconductor industry structure and application areas | 1.1 IC Design<br>1.2 Wafer Fabrication<br>1.3 Assembly and Packaging<br>1.4 Testing<br>1.5 Consumer Electronics<br>1.6 Automotive and Industrial Applications |
| 2. Design models                                          | 2.1 ASIC<br>2.2 FPGA<br>2.3 SoC<br>2.4 Microcontroller (MCU)<br>2.5 System-in-Package (SiP)<br>2.6 Application-Specific Standard Product (ASSP)               |
| 3. CMOS technology and logic gate operations              | 3.1 CMOS Technology<br>3.2 NOT Gate<br>3.3 AND Gate<br>3.4 OR Gate<br>3.5 NAND Gate<br>3.6 NOR Gate                                                           |
| 4. Basic timing concepts                                  | 4.1 Clock Signal<br>4.2 Setup Time<br>4.3 Hold Time<br>4.4 Propagation Delay<br>4.5 Clock Skew<br>4.6 Clock Jitter                                            |
| 5. Timing parameters                                      | 5.1 Setup Time<br>5.2 Hold Time<br>5.3 Clock Period<br>5.4 Propagation Delay<br>5.5 Clock-to-Q Delay<br>5.6 Slack                                             |
| 6. CMOS design principles                                 | 6.1 CMOS Inverter<br>6.2 Pull-Up Network (PUN)<br>6.3 Pull-Down Network (PDN)<br>6.4 Power Consumption<br>6.5 Noise Margin<br>6.6 Fan-In/Fan-Out              |
| 7. ASIC design flow                                       | 7.1 RTL Design<br>7.2 Functional Verification<br>7.3 Logic Synthesis<br>7.4 Physical Design<br>7.5 Static Timing Analysis (STA)                               |

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|                                      | 7.6 GDSII Generation                                                                                                                         |
| 8. EDA tools and design environments | 8.1 Cadence Innovus<br>8.2 Cadence Genus<br>8.3 Synopsys Design Compiler<br>8.4 Synopsys PrimeTime<br>8.5 Siemens Questa<br>8.6 Synopsys VCS |

## Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Semiconductor industry structure, manufacturing processes, application areas, and career opportunities.<br>1.2 Fundamentals of semiconductor design, fabrication, packaging, testing, and design business models (Fabless, Foundry, IDM).<br>1.3 CMOS technology, digital logic gates, and basic digital circuit design principles.<br>1.4 Timing concepts and timing analysis.<br>1.5 Complete ASIC design flow from RTL to GDSII and back-end implementation stages.<br>1.6 Design files, standard cell libraries, technology libraries, design constraints, and timing requirements.<br>1.7 Functions and applications of Electronic Design Automation (EDA) tools in ASIC physical design. |
| 2. Underpinning Skills    | 2.1 Applying knowledge of semiconductor industry structure, processes, and career pathways<br>2.2 Applying CMOS technology and digital logic design fundamentals<br>2.3 Applying CMOS design principles to basic digital circuits<br>2.4 Interpreting the ASIC design flow from RTL to GDSII<br>2.5 Identifying design files, technology libraries, constraints, and EDA tools<br>2.6 Analyzing design reports, timing data, and simulation results for verification<br>2.7 Using ASIC design data to support implementation and quality assurance                                                                                                                                                 |
| 3 Underpinning Attitudes  | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincere and honest to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Communication with peers, subordinates and seniors in the workplace                                                                                                                                                                                                                                                                                                                                |

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| 4 Resource Implications | 4.1 Workplace (simulated or actual)<br>4.2 Projector<br>4.3 Stationery<br>4.4 Learning manual |
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### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                               |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 described semiconductor industry and design models<br>1.2 applied CMOS and timing fundamentals<br>1.3 interpreted ASIC design flow and design data                                                    |
| 2 Methods of Assessment           | Competency should be assessed by:<br>2.1 Written test<br>2.2 Practical Demonstration<br>2.3 Oral Questioning<br>2.4 Portfolio (Optional)                                                                                                                      |
| 3 Context of Assessment           | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert. |

|                                                                                                                                                                                                                                                                                                                   |                                     |                                          |
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| <b>Unit of Competency:</b><br><b>OPERATE LINUX, EDA TOOLS AND AUTOMATION SCRIPTS</b>                                                                                                                                                                                                                              | <b>Nominal Duration:</b><br>16 hrs. | <b>Unit Code:</b><br>SICIP-SCD-APDP-02-S |
| <b>Unit Descriptor:</b><br><br>This unit covers the knowledge, skills and attitudes required to Operate LINUX, EDA tools and automation scripts. It specifically includes the tasks of operate Linux EDA tools, applying shell and TCL scripting, using version control systems and executing automation scripts. |                                     |                                          |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency           | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Operate Linux EDA tools       | 1.1 <b><u>Linux operating system</u></b> is used to manage files, directories, and project data.<br>1.2 Linux commands are executed to perform development and administration tasks.<br>1.3 <b><u>EDA tools</u></b> are launched and configured according to project requirements.<br>1.4 Design files and project directories are managed within the EDA environment.<br>1.5 <b><u>Tool execution and output logs</u></b> are reviewed to verify successful operation. |
| 2. Apply shell and TCL scripting | 2.1 <b><u>Shell scripts</u></b> are created to automate routine Linux tasks.<br>2.2 <b><u>TCL scripts</u></b> are developed to automate EDA tool operations.<br>2.3 Script variables, procedures, and control statements are applied according to task requirements.<br>2.4 Scripts are executed and tested to verify expected results.<br>2.5 <b><u>Script errors</u></b> are identified and corrected through debugging.                                              |
| 3. Use version control systems   | 3.1 <b><u>Version control repositories</u></b> are created and configured for design projects.<br>3.2 Design files and scripts are managed using version control commands.<br>3.3 Changes are committed and documented using <b><u>appropriate commit messages</u></b> .<br>3.4 Branches are created, merged, and synchronized according to project requirements.<br>3.5 Project versions are maintained to support collaboration and design tracking.                  |
| 4. Execute automation scripts    | 4.1 <b><u>Automation scripts</u></b> are executed according to project requirements.<br>4.2 Script inputs and execution parameters are configured                                                                                                                                                                                                                                                                                                                       |

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|  | correctly.                                                                           |
|  | 4.3 Automated tasks are monitored using execution logs and reports.                  |
|  | 4.4 <b><u>Script execution errors</u></b> are identified and resolved.               |
|  | 4.5 Automation results are verified to ensure successful completion of design tasks. |

## Range of Variables

| Variable                          | Range                                                                                                                                                             |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | May include but not limited to:                                                                                                                                   |
| 1. Linux operating system         | 1.1 Ubuntu<br>1.2 Red Hat Enterprise Linux (RHEL)<br>1.3 CentOS<br>1.4 Rocky Linux<br>1.5 Oracle Linux<br>1.6 SUSE Linux                                          |
| 2. EDA tools                      | 2.1 Cadence Innovus<br>2.2 Cadence Genus<br>2.3 Synopsys Design Compiler<br>2.4 Synopsys PrimeTime<br>2.5 Siemens Questa<br>2.6 Synopsys VCS                      |
| 3. Tool execution and output logs | 3.1 Compilation Logs<br>3.2 Simulation Logs<br>3.3 Synthesis Logs<br>3.4 Timing Reports<br>3.5 Error Logs<br>3.6 Warning Logs                                     |
| 4. Shell scripts                  | 4.1 Bash Scripts<br>4.2 File Management Scripts<br>4.3 Backup Scripts<br>4.4 Build Scripts<br>4.5 Automation Scripts<br>4.6 Execution Scripts                     |
| 5. TCL scripts                    | 5.1 Tool Setup Scripts<br>5.2 Synthesis Scripts<br>5.3 Simulation Scripts<br>5.4 Physical Design Scripts<br>5.5 Timing Analysis Scripts<br>5.6 Automation Scripts |
| 6. Script errors                  | 6.1 Syntax Errors<br>6.2 Runtime Errors                                                                                                                           |

|                                 |                                                                                                                                                      |
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|                                 | 6.3 Logic Errors<br>6.4 Debug Messages<br>6.5 Error Logs<br>6.6 Code Correction                                                                      |
| 7. Version control repositories | 7.1 Git Repository<br>7.2 GitHub Repository<br>7.3 GitLab Repository<br>7.4 Bitbucket Repository<br>7.5 Local Repository<br>7.6 Remote Repository    |
| 8. Appropriate commit messages  | 8.1 Initial Commit<br>8.2 Feature Update<br>8.3 Bug Fix<br>8.4 Code Refactoring<br>8.5 Documentation Update<br>8.6 Version Release                   |
| 9. Automation scripts           | 9.1 Shell Scripts<br>9.2 TCL Scripts<br>9.3 Python Scripts<br>9.4 Build Scripts<br>9.5 Flow Automation Scripts<br>9.6 Batch Scripts                  |
| 10. Script execution errors     | 10.1 Syntax Errors<br>10.2 Runtime Errors<br>10.3 Configuration Errors<br>10.4 File Access Errors<br>10.5 Tool Execution Errors<br>10.6 Logic Errors |

## Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning knowledge | 1.1 Fundamentals of the Linux operating system, file system structure, and command-line operations.<br>1.2 Linux file, directory, user, and permission management for EDA project environments.<br>1.3 Principles of Electronic Design Automation (EDA) tool setup, configuration, and project management.<br>1.4 Fundamentals of Shell scripting and TCL scripting, including variables, procedures, and control structures.<br>1.5 Techniques for script execution, testing, debugging, and automation of repetitive EDA tasks.<br>1.6 Principles of version control systems and commit practices. |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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|                           | 1.7 Methods for executing automation scripts, configuring parameters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2. Underpinning skills    | 2.1 Operating Linux environments to manage project files and EDA workflows<br>2.2 Using Linux commands for development, administration, and project management<br>2.3 Configuring and operating EDA tools according to project requirements<br>2.4 Applying Shell and TCL scripting to automate design and system tasks<br>2.5 Developing, testing, and debugging automation scripts for EDA operations<br>2.6 Maintaining project versions through commits, branches, and collaboration practices<br>2.7 Executing, monitoring, and verifying automation scripts to ensure successful design operations |
| 3. Underpinning attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincerity and honesty to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Good relationships with peers, sub-ordinates and seniors in the workplace                                                                                                                                                                                                                             |
| 4. Resource implications  | The following resources must be provided:<br>4.1 Workplace (simulated or actual)<br>4.2 Learning manual<br>4.3 EDA tools<br>4.4 Project files                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                            |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical aspects of competency | Assessment must evidence that the candidate:<br>1.1 operated Linux EDA tools<br>1.2 applied shell and TCL scripting<br>1.3 used version control systems<br>1.4 executed automation scripts |
| 2. Methods of assessment          | Methods of assessment may include but is not limited to:<br>2.1 Written test<br>2.2 Practical demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (Optional)                            |



|                          |                                                                                                                                                                                                                                                                          |
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| 3. Context of assessment | <p>3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</p> <p>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</p> |
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## Occupation-specific Competencies

|                                                                                                                                                                                                                                                                                                                                                                                                                             |                                     |                                          |
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| <b>Unit of Competency:</b><br><b>DEVELOP SYNTHESIZABLE RTL USING VERILOG/SYSTEM VERILOG</b>                                                                                                                                                                                                                                                                                                                                 | <b>Nominal Duration:</b><br>32 hrs. | <b>Unit Code:</b><br>SICIP-SCD-APDP-01-O |
| <b>Unit Descriptor:</b><br><br>This unit covers the knowledge, skills and attitudes required to develop synthesizable RTL using Verilog/system Verilog. It specifically includes the tasks of implementing combinational logic, implementing sequential and FSM designs, applying RTL coding guidelines, preparing testbench and simulation setup, running simulation and debug waveforms and validating RTL functionality. |                                     |                                          |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                   | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Implement combinational logic.        | 1.1 <b><u>Combinational logic requirements</u></b> are analyzed from the design specification.<br>1.2 <b><u>Verilog/SystemVerilog modules</u></b> are developed using synthesizable coding constructs.<br>1.3 Logic operators, conditional statements, and continuous assignments are applied.<br>1.4 RTL code is compiled and checked for syntax errors.<br>1.5 <b><u>Functional simulation</u></b> is performed to verify combinational logic behavior.                            |
| 2. Implement sequential and FSM designs. | 2.1 <b><u>Sequential logic and FSM requirements</u></b> are interpreted from the design specification.<br>2.2 Registers, counters, and flip-flops are implemented using <b><u>synthesizable RTL code</u></b> .<br>2.3 FSM states, transitions, and output logic are developed according to design requirements.<br>2.4 <b><u>Clock and reset logic</u></b> are implemented following RTL design practices.<br>2.5 Sequential circuits are simulated to verify correct functionality. |
| 3. Apply RTL coding guidelines.          | 3.1 RTL code is written following <b><u>synthesizable coding standards</u></b> .<br>3.2 Meaningful module names, signals, and comments are applied for readability.<br>3.3 <b><u>Coding practices</u></b> are applied to avoid latches, race conditions, and synthesis issues.<br>3.4 RTL code is reviewed for quality, consistency, and maintainability.<br>3.5 Coding guideline violations are identified and corrected.                                                           |
| 4. Prepare testbench                     | 4.1 <b><u>Testbench</u></b> is developed according to design                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                  |
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| and simulation setup.                  | <p>requirements.</p> <p>4.2 Test vectors and input stimulus are prepared for functional verification.</p> <p>4.3 <b><u>Simulation environment</u></b> is configured using appropriate EDA tools.</p> <p>4.4 Design and testbench files are compiled before simulation.</p> <p>4.5 Simulation setup is verified to ensure correct execution.</p>                                                  |
| 5. Run simulation and debug waveforms. | <p>5.1 RTL simulation is executed using appropriate <b><u>RTL simulation tools</u></b>.</p> <p>5.2 Simulation waveforms are analyzed to verify signal behavior.</p> <p>5.3 <b><u>Functional errors</u></b> are identified through waveform analysis.</p> <p>5.4 RTL code is modified to resolve simulation issues.</p> <p>5.5 Simulation results are verified against design specifications.</p> |
| 6. Validate RTL functionality.         | <p>6.1 <b><u>RTL functionality</u></b> is verified against the design specification.</p> <p>6.2 Functional outputs are compared with expected results.</p> <p>6.3 Verification results are documented and reviewed.</p> <p>6.4 Identified functional issues are corrected before design completion.</p> <p>6.5 <b><u>Validated RTL design</u></b> is prepared for synthesis.</p>                 |

### Range of Variables

| Variable                            | Range<br>May include but not limited to:                                                                                                                   |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Combinational logic requirements | <p>1.1 Logic Gates</p> <p>1.2 Multiplexers</p> <p>1.3 Decoders</p> <p>1.4 Encoders</p> <p>1.5 Comparators</p> <p>1.6 Arithmetic Logic Units (ALUs)</p>     |
| 2. Verilog/SystemVerilog modules    | <p>2.1 Module</p> <p>2.2 Input/Output Ports</p> <p>2.3 Wire</p> <p>2.4 Assign Statement</p> <p>2.5 Always_comb Block</p> <p>2.6 Parameter</p>              |
| 3. Functional simulation            | <p>3.1 Testbench</p> <p>3.2 Waveforms</p> <p>3.3 Simulation Results</p> <p>3.4 Input Stimuli</p> <p>3.5 Output Verification</p> <p>3.6 Simulation Logs</p> |
| 4. Sequential logic and             | <p>4.1 Registers</p>                                                                                                                                       |

|                                   |                                                                                                                                                                              |
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| FSM requirements                  | 4.2 Counters<br>4.3 Flip-Flops<br>4.4 Finite State Machines (FSM)<br>4.5 Shift Registers<br>4.6 Control Logic                                                                |
| 5. Synthesizable RTL code         | 5.1 D Flip-Flop<br>5.2 Register<br>5.3 Counter<br>5.4 Shift Register<br>5.5 Always_ff Block<br>5.6 Non-Blocking Assignment (<=)                                              |
| 6. Clock and reset logic          | 6.1 Clock Signal<br>6.2 Synchronous Reset<br>6.3 Asynchronous Reset<br>6.4 Clock Enable<br>6.5 Rising Edge Trigger<br>6.6 Falling Edge Trigger                               |
| 7. Synthesizable coding standards | 7.1 Verilog Coding Standards<br>7.2 SystemVerilog Coding Standards<br>7.3 Synthesizable Constructs<br>7.4 RTL Coding Rules<br>7.5 Naming Conventions<br>7.6 Coding Templates |
| 8. Coding practices               | 8.1 Avoid Latches<br>8.2 Avoid Race Conditions<br>8.3 Non-Blocking Assignments<br>8.4 Blocking Assignments<br>8.5 Reset Logic<br>8.6 Clocked Processes                       |
| 9. Testbench                      | 9.1 Testbench Module<br>9.2 DUT (Design Under Test)<br>9.3 Clock Generator<br>9.4 Reset Generator<br>9.5 Stimulus Generator<br>9.6 Monitor                                   |
| 10. Simulation environment        | 10.1 Synopsys VCS<br>10.2 Siemens Questa<br>10.3 Cadence Xcelium<br>10.4 Simulation Libraries<br>10.5 Simulation Scripts<br>10.6 Project Workspace                           |
| 11. RTL simulation tools          | 11.1 Synopsys VCS<br>11.2 Siemens Questa<br>11.3 Cadence Xcelium                                                                                                             |

|                          |                                                                                                                                                               |
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|                          | 11.4 Verilator<br>11.5 ModelSim<br>11.6 Vivado Simulator                                                                                                      |
| 12. Functional errors    | 12.1 Logic Errors<br>12.2 Timing Errors<br>12.3 State Transition Errors<br>12.4 Reset Errors<br>12.5 Output Mismatch<br>12.6 Assertion Failures               |
| 13. RTL functionality    | 13.1 RTL Specification<br>13.2 Functional Requirements<br>13.3 Design Behavior<br>13.4 Input/Output Logic<br>13.5 State Machines<br>13.6 Control Logic        |
| 14. Validated RTL design | 14.1 Synthesizable RTL<br>14.2 RTL Source Files<br>14.3 Constraint Files<br>14.4 Design Documentation<br>14.5 Verified Netlist<br>14.6 Synthesis-Ready Design |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning knowledge | 1.1 Fundamentals of digital logic design and finite state machines (FSMs).<br>1.2 RTL design principles.<br>1.3 RTL coding guidelines, naming conventions, and best practices.<br>1.4 Principles of clocking, reset design, registers, counters, and FSM implementation.<br>1.5 Testbench development, stimulus generation, and functional simulation concepts.<br>1.6 Use of RTL simulation tools.<br>1.7 Interpretation of simulation results, verification against design specifications. |
| 2. Underpinning skills    | 2.1 Applying RTL design principles to implement combinational, sequential, and FSM circuits<br>2.2 Developing synthesizable Verilog/SystemVerilog code following coding standards<br>2.3 Applying RTL coding guidelines to improve readability, quality, and maintainability<br>2.4 Preparing testbenches, test vectors, and simulation environments for functional verification                                                                                                             |

|                           |                                                                                                                                                                                                                                                                                                                                                                              |
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|                           | 2.5 Validating RTL functionality against design specifications and expected outputs<br>2.6 Preparing verified RTL designs for synthesis and subsequent implementation stages                                                                                                                                                                                                 |
| 3. Underpinning attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincerity and honesty to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Good relationships with peers, sub-ordinates and seniors in the workplace |
| 4. Resource implications  | The following resources must be provided:<br><br>4.1 Workplace (simulated or actual)<br>4.2 Learning materials<br>4.3 Tools & equipment<br>4.4 Job specifications and work instructions                                                                                                                                                                                      |

#### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                          |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical aspects of competency | Assessment must evidence that the candidate:<br>1.1 implemented combinational logic.<br>1.2 implemented sequential and FSM designs.<br>1.3 applied RTL coding guidelines.<br>1.4 prepared testbench and simulation setup.<br>1.5 run simulation and debug waveforms.<br>1.6 validated RTL functionality. |
| 2. Methods of assessment          | Methods of assessment should include:<br>2.1 Written Test<br>2.2 Practical demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (optional)                                                                                                                                                             |
| 3. Context of assessment          | 3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.                                           |

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| <b>Unit of Competency:</b><br><b>PERFORM LOGIC SYNTHESIS AND TIMING ANALYSIS</b>                                                                                                                                                                                                                                                                                                                                         | <b>Nominal Duration:</b><br>36 hrs. | <b>Unit Code:</b><br>SICIP-SCD-APDP-02-O |
| <b>Unit Descriptor:</b><br><br>This unit of competency covers the knowledge, skills and attitudes required to perform logic synthesis and timing analysis. It specifically includes the tasks of executing logic synthesis flow, creating and debugging timing constraints, optimizing and analyzing QoR, analyzing timing fundamentals, performing static timing analysis (STA) and applying timing closure techniques. |                                     |                                          |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                  | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Execute logic synthesis flow.        | 1.1 <b><u>RTL design and technology libraries</u></b> are prepared for synthesis.<br>1.2 Synthesis constraints and input files are configured according to design requirements.<br>1.3 Logic synthesis is executed using the appropriate EDA tool.<br>1.4 Synthesis reports and generated netlists are reviewed for correctness.<br>1.5 <b><u>Synthesis issues</u></b> are identified and corrected before implementation.    |
| 2. Create and debug timing constraints. | 2.1 <b><u>Timing constraints</u></b> are created according to design specifications.<br>2.2 Clock definitions, input delays, and output delays are applied correctly.<br>2.3 Timing constraint files are verified for completeness and accuracy.<br>2.4 <b><u>Constraint violations and errors</u></b> are identified through timing reports.<br>2.5 Timing constraints are updated and validated to ensure correct analysis. |
| 3. Optimize and analyze QoR.            | 3.1 <b><u>QoR (Quality of Results) reports</u></b> are generated after synthesis.<br>3.2 Area, timing, and power reports are analyzed according to design goals.<br>3.3 <b><u>Design optimization techniques</u></b> are applied to improve QoR.<br>3.4 Optimization results are verified against project requirements.<br>3.5 Final QoR reports are reviewed and documented.                                                 |

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| 4. Analyze timing fundamentals.          | <p>4.1 <b><u>Timing paths and clock relationships</u></b> are identified from the design.</p> <p>4.2 Setup time, hold time, clock skew, and clock latency are analyzed.</p> <p>4.3 <b><u>Critical paths and timing violations</u></b> are identified from timing reports.</p> <p>4.4 Timing calculations are interpreted according to design requirements.</p> <p>4.5 Timing analysis results are reviewed before optimization.</p> |
| 5. Perform static timing analysis (STA). | <p>5.1 Static Timing Analysis (STA) is performed using the appropriate EDA tool.</p> <p>5.2 Timing reports are generated and analyzed for design verification.</p> <p>5.3 Setup and hold violations are identified from STA reports.</p> <p>5.4 <b><u>Timing exceptions</u></b> are verified and applied where required.</p> <p>5.5 STA results are validated to ensure timing requirements are achieved.</p>                       |
| 6. Apply timing closure techniques.      | <p>6.1 <b><u>Timing violations</u></b> are identified from synthesis and STA reports.</p> <p>6.2 Timing optimization techniques are applied to improve design performance.</p> <p>6.3 Design constraints are refined to achieve timing closure.</p> <p>6.4 Timing analysis is repeated to verify optimization results.</p> <p>6.5 Final design is validated after successful timing closure.</p>                                    |

### Range of Variables

| Variable                               | Range                                                                                                                                                                              |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | May include but not limited to:                                                                                                                                                    |
| 1. RTL design and technology libraries | <p>1.1 RTL Source Files</p> <p>1.2 Standard Cell Libraries</p> <p>1.3 Technology Libraries</p> <p>1.4 Liberty (.lib) Files</p> <p>1.5 Design Files</p> <p>1.6 Constraint Files</p> |
| 2. Synthesis issues                    | <p>2.1 Timing Violations</p> <p>2.2 Unmapped Logic</p> <p>2.3 Constraint Violations</p> <p>2.4 Library Errors</p> <p>2.5 RTL Coding Issues</p> <p>2.6 Optimization Warnings</p>    |
| 3. Timing constraints                  | <p>3.1 SDC File</p> <p>3.2 Clock Constraints</p> <p>3.3 Input Delay Constraints</p> <p>3.4 Output Delay Constraints</p> <p>3.5 Timing Exceptions</p>                               |



|                                         |                                                                                                                                                          |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         | 3.6 Design Constraints                                                                                                                                   |
| 4. Constraint violations and errors     | 4.1 Setup Violations<br>4.2 Hold Violations<br>4.3 Unconstrained Paths<br>4.4 Clock Definition Errors<br>4.5 Constraint Conflicts<br>4.6 Timing Warnings |
| 5. QoR (Quality of Results) reports     | 5.1 Area Report<br>5.2 Timing Report<br>5.3 Power Report<br>5.4 QoR Summary Report<br>5.5 Utilization Report<br>5.6 Optimization Report                  |
| 6. Design optimization techniques       | 6.1 Area Optimization<br>6.2 Timing Optimization<br>6.3 Power Optimization<br>6.4 Logic Optimization<br>6.5 Gate Sizing<br>6.6 Buffer Insertion          |
| 7. Timing paths and clock relationships | 7.1 Data Path<br>7.2 Clock Path<br>7.3 Launch Clock<br>7.4 Capture Clock<br>7.5 Critical Path<br>7.6 Clock Domain                                        |
| 8. Critical paths and timing violations | 8.1 Critical Path<br>8.2 Setup Violation<br>8.3 Hold Violation<br>8.4 Slack Violation<br>8.5 Unconstrained Path<br>8.6 Timing Exceptions                 |
| 9. Timing exceptions                    | 9.1 False Path<br>9.2 Multi-Cycle Path<br>9.3 Clock Groups<br>9.4 Input Delay Exception<br>9.5 Output Delay Exception<br>9.6 Disable Timing              |
| 10. Timing violations                   | 10.1 Setup Violations<br>10.2 Hold Violations<br>10.3 Negative Slack<br>10.4 Critical Paths<br>10.5 Clock Violations<br>10.6 Constraint Violations       |

## Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning knowledge | <ul style="list-style-type: none"> <li>1.1 Principles of logic synthesis, technology libraries, gate-level netlist generation.</li> <li>1.2 Timing constraint development using SDC and timing exceptions.</li> <li>1.3 Fundamentals of Quality of Results (QoR), including area, timing, power, and optimization trade-offs.</li> <li>1.4 Timing analysis concepts and critical paths.</li> <li>1.5 Principles and methodology of Static Timing Analysis (STA) and timing report interpretation.</li> <li>1.6 Techniques for identifying and resolving timing violations through optimization and constraint refinement.</li> <li>1.7 Timing closure methodologies, iterative optimization, and final design validation before physical implementation.</li> </ul> |
| 2. Underpinning skills    | <ul style="list-style-type: none"> <li>2.1 Executing logic synthesis using RTL designs, technology libraries, and EDA tools</li> <li>2.2 Creating, verifying, and debugging timing constraints for accurate analysis</li> <li>2.3 Analyzing and optimizing Quality of Results (QoR) for area, timing, and power</li> <li>2.4 Applying timing fundamentals including setup, hold, clock skew, and latency analysis</li> <li>2.5 Performing Static Timing Analysis (STA) and interpreting timing reports</li> <li>2.6 Applying timing optimization and timing closure techniques to improve design performance</li> <li>2.7 Reviewing synthesis, STA, and QoR reports to ensure design readiness for physical implementation</li> </ul>                               |
| 3. Underpinning attitudes | <ul style="list-style-type: none"> <li>3.1 Commitment to occupational health and safety</li> <li>3.2 Promptness in carrying out activities</li> <li>3.3 Sincerity and honesty to duties</li> <li>3.4 Environmental concerns</li> <li>3.5 Eagerness to learn</li> <li>3.6 Tidiness and timeliness</li> <li>3.7 Respect for rights of peers and seniors in the workplace</li> <li>3.8 Good relationships with peers, sub-ordinates and seniors in the workplace</li> </ul>                                                                                                                                                                                                                                                                                            |
| 4. Resource implications  | <p>The following resources must be provided:</p> <ul style="list-style-type: none"> <li>4.1 Workplace (simulated or actual)</li> <li>4.2 Specifications and work instructions</li> <li>4.3 Learning materials</li> <li>4.4 Tools and equipment</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical aspects of competency | <p>Assessment must evidence that the candidate:</p> <ul style="list-style-type: none"><li>1.1 executed logic synthesis flow.</li><li>1.2 created and debug timing constraints.</li><li>1.3 optimized and analyzed QoR.</li><li>1.4 analyzed timing fundamentals.</li><li>1.5 performed static timing analysis (STA).</li><li>1.6 applied timing closure techniques.</li></ul> |
| 2. Methods of assessment          | <p>Methods of assessment should include:</p> <ul style="list-style-type: none"><li>2.1 Written test</li><li>2.2 Practical demonstration</li><li>2.3 Oral questioning</li><li>2.4 Portfolio (optional)</li></ul>                                                                                                                                                               |
| 3. Context of assessment          | <ul style="list-style-type: none"><li>3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training.</li><li>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</li></ul>                                                           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                     |                                          |
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| <b>Unit of Competency:</b><br><b>CARRY OUT PHYSICAL DESIGN IMPLEMENTATION</b>                                                                                                                                                                                                                                                                                                                                                                     | <b>Nominal Duration:</b><br>52 hrs. | <b>Unit Code:</b><br>SICIP-SCD-APDP-03-O |
| <b>Unit Descriptor:</b><br><br>This unit covers the knowledge, skills and attitudes required to carry out physical design implementation. It specifically includes the tasks of performing floor planning and power planning, performing placement and congestion analysis, performing clock tree synthesis (CTS), performing routing and physical optimization, performing extraction and parasitic analysis and preparing final GDSII database. |                                     |                                          |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                        | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Perform floor planning and power planning. | 1.1 Design netlist, technology libraries, and floorplan constraints are prepared.<br>1.2 Core area, I/O placement, and macro locations are defined according to design requirements.<br>1.3 Power rings, power straps, and <b><u>power distribution network</u></b> are created.<br>1.4 Floorplan is verified for area utilization and design feasibility.<br>1.5 Floor planning and power planning reports are reviewed before placement. |
| 2. Perform placement and congestion analysis. | 2.1 Standard cells are placed according to floorplan constraints.<br>2.2 <b><u>Placement quality</u></b> is analyzed for utilization and timing requirements.<br>2.3 <b><u>Congestion hotspots</u></b> are identified using placement reports.<br>2.4 Placement optimization is performed to reduce congestion and improve timing.<br>2.5 Placement results are verified before clock tree synthesis.                                      |
| 3. Perform clock tree synthesis (CTS).        | 3.1 <b><u>Clock tree synthesis</u></b> is configured according to design constraints.<br>3.2 Clock networks are generated using the appropriate EDA tool.<br>3.3 Clock skew, latency, and insertion delay are analyzed.<br>3.4 <b><u>CTS optimization</u></b> is performed to improve clock distribution.<br>3.5 CTS reports are reviewed and validated before routing.                                                                    |
| 4. Perform routing and physical optimization. | 4.1 Global and detailed routing are performed according to design requirements.<br>4.2 Routing violations and design rule issues are identified                                                                                                                                                                                                                                                                                            |

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | <p>and corrected.</p> <p>4.3 <b><u>Physical optimization techniques</u></b> are applied to improve timing and routing quality.</p> <p>4.4 Post-routing timing and design quality are verified.</p> <p>4.5 Routing reports are reviewed before sign-off analysis.</p>                                                                                                                                                      |
| 5. Perform extraction and parasitic analysis. | <p>5.1 RC extraction is performed using the appropriate EDA tool.</p> <p>5.2 <b><u>Parasitic data</u></b> is generated and verified for accuracy.</p> <p>5.3 Extracted data is analyzed to evaluate timing impact.</p> <p>5.4 Timing is revalidated using extracted parasitic information.</p> <p>5.5 Extraction reports are reviewed before final database generation.</p>                                               |
| 6. Prepare final GDSII database.              | <p>6.1 <b><u>Final physical design</u></b> is verified for design rule and layout compliance.</p> <p>6.2 Sign-off checks are completed according to project requirements.</p> <p>6.3 GDSII database is generated using the approved design flow.</p> <p>6.4 Generated GDSII files and sign-off reports are verified for completeness.</p> <p>6.5 Final design database is prepared for tape-out and project delivery.</p> |

### Range of Variables

| Variable                      | Range                                                                                                                                                          |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | May include but not limited to:                                                                                                                                |
| 1. Power distribution network | <p>1.1 Power Rings</p> <p>1.2 Power Straps</p> <p>1.3 Power Grid</p> <p>1.4 VDD Network</p> <p>1.5 VSS Network</p> <p>1.6 Power Pads</p>                       |
| 2. Placement quality          | <p>2.1 Area Utilization</p> <p>2.2 Timing Performance</p> <p>2.3 Cell Density</p> <p>2.4 Wirelength</p> <p>2.5 Placement Efficiency</p> <p>2.6 QoR Metrics</p> |
| 3. Congestion hotspots        | <p>3.1 Routing Congestion</p> <p>3.2 Congestion Map</p> <p>3.3 High-Density Areas</p> <p>3.4 Overflow Regions</p> <p>3.5 Placement Reports</p>                 |

|                                     |                                                                                                                                                          |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | 3.6 Congestion Reports                                                                                                                                   |
| 4. Clock tree synthesis             | 4.1 Clock Constraints<br>4.2 CTS Constraints<br>4.3 Clock Buffers<br>4.4 Clock Gates<br>4.5 Clock Tree Specification<br>4.6 Skew Targets                 |
| 5. CTS optimization                 | 5.1 Buffer Insertion<br>5.2 Clock Balancing<br>5.3 Skew Optimization<br>5.4 Latency Optimization<br>5.5 Transition Optimization<br>5.6 Clock Tree Repair |
| 6. Physical optimization techniques | 6.1 Timing Optimization<br>6.2 Buffer Insertion<br>6.3 Cell Resizing<br>6.4 Routing Optimization<br>6.5 Congestion Optimization<br>6.6 ECO Optimization  |
| 7. Parasitic data                   | 7.1 SPEF File<br>7.2 DSPF File<br>7.3 RSPF File<br>7.4 Resistance Data<br>7.5 Capacitance Data<br>7.6 Coupling Capacitance                               |
| 8. Final physical design            | 8.1 Design Rule Check (DRC)<br>8.2 Layout Versus Schematic (LVS)<br>8.3 Antenna Check<br>8.4 Density Check<br>8.5 Physical Verification                  |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning knowledge | 1.1 Fundamentals of physical design implementation.<br>1.2 Principles of floorplanning, macro placement, I/O planning.<br>1.3 Cell placement methodologies and timing-driven placement techniques.<br>1.4 Clock Tree Synthesis (CTS) concepts and clock optimization.<br>1.5 Routing techniques, physical optimization, design rule checking (DRC).<br>1.6 Physical verification, sign-off checks, DRC/LVS compliance, and GDSII generation for tape-out. |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | 1.7 Interpretation of implementation reports, sign-off reports.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2. Underpinning skills    | 2.1 Applying floor planning and power planning techniques for physical design implementation<br>2.2 Performing standard cell placement and congestion analysis to optimize layout quality<br>2.3 Performing RC extraction and parasitic analysis to validate post-layout timing<br>2.4 Analyzing timing, congestion, and routing reports to identify and resolve design issues<br>2.5 Applying physical design optimization techniques to achieve timing closure and design rule compliance<br>2.6 Reviewing physical design reports and preparing the final database for project delivery |
| 3. Underpinning attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincerity and honesty to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in workplace<br>3.8 Good relationships with peers, sub-ordinates and seniors in workplace                                                                                                                                                                                                                       |
| 4. Resource implications  | <p>The Following resources must be provided:</p> 4.1 Workplace (simulated or actual)<br>4.2 Learning manual<br>4.3 Job specifications and work instructions                                                                                                                                                                                                                                                                                                                                                                                                                                |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical aspects of competency | <p>Assessment must evidence that the candidate:</p> 1.1 performed floor planning and power planning.<br>1.2 performed placement and congestion analysis.<br>1.3 performed clock tree synthesis (CTS).<br>1.4 performed routing and physical optimization.<br>1.5 performed extraction and parasitic analysis.<br>1.6 prepared final GDSII database. |
| 2. Methods of assessment          | <p>Methods of assessment should include:</p> 2.1 Written test<br>2.2 Practical demonstration<br>2.3 Oral questioning<br>2.4 Portfolio (optional)                                                                                                                                                                                                    |
| 3. Context of assessment          | 3.1 Competency assessment must be done in an assessment/ training center or in an actual or                                                                                                                                                                                                                                                         |

|  |                                                                                                                                                                               |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>simulated work place after completion of the training.</p> <p>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</p> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------|
| <b>Unit of Competency:</b><br><b>PERFORM PHYSICAL VERIFICATION AND SIGNOFF</b>                                                                                                                                                                                                                                                                                       | <b>Nominal Duration:</b><br>24 hrs. | <b>Unit Code:</b><br>SICIP-SCD-APDP-04-O |
| <b>Unit Descriptor:</b><br><br>This unit of competency covers the knowledge, skills and attitudes required to perform physical verification and signoff. It specifically includes the tasks of performing DRC and LVS verification, performing signoff checks, analyzing power components, analyzing signal integrity and reliability and reviewing signoff results. |                                     |                                          |

**Elements and Performance Criteria:**

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                       | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Perform DRC and LVS verification.         | 1.1 <b><u>Design rule checking (DRC)</u></b> is performed using the appropriate physical verification tool.<br>1.2 Layout Versus Schematic (LVS) verification is executed to confirm layout accuracy.<br>1.3 DRC and LVS violations are identified from verification reports.<br>1.4 Verification errors are analyzed and corrected according to design requirements.<br>1.5 DRC and LVS results are validated before signoff. |
| 4. Perform signoff checks.                   | 2.1 <b><u>Signoff verification flow</u></b> is executed according to project requirements.<br>2.2 Timing, physical verification, and reliability reports are reviewed.<br>2.3 <b><u>Signoff criteria</u></b> are verified against design specifications.<br>2.4 Outstanding violations are identified and resolved before signoff.<br>2.5 Final signoff status is confirmed for tape-out readiness.                            |
| 3. Analyze power components.                 | 3.1 Dynamic and leakage power reports are generated using the appropriate EDA tool.<br>3.2 <b><u>Power consumption and distribution</u></b> are analyzed against design targets.<br>3.3 IR drop and power integrity issues are identified and evaluated.<br>3.4 <b><u>Power optimization opportunities</u></b> are identified based on analysis results.<br>3.5 Power analysis results are verified before final signoff.      |
| 4. Analyze signal integrity and reliability. | 4.1 <b><u>Signal integrity analysis</u></b> is performed to identify crosstalk and noise issues.<br>4.2 Reliability checks are conducted for electromigration (EM) and IR drop.<br>4.3 Signal integrity and reliability violations are analyzed                                                                                                                                                                                |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | <p>and documented.</p> <p>4.4 <b><u>Corrective actions</u></b> are applied to improve design reliability.</p> <p>4.5 Analysis results are verified to ensure signoff requirements are achieved.</p>                                                                                                                                                                                                                          |
| 5. Review signoff results. | <p>5.1 <b><u>Signoff reports</u></b> are reviewed for completeness and accuracy.</p> <p>5.2 Design quality is evaluated against project specifications and signoff criteria.</p> <p>5.3 Outstanding issues are identified and resolved before design release.</p> <p>5.4 Final verification reports are documented and approved.</p> <p>5.5 <b><u>Signoff package</u></b> is prepared for tape-out and project delivery.</p> |

### Range of Variables

| Variable                              | Range<br>May include but not limited to:                                                                                                                              |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Design rule checking (DRC)         | <p>1.1 DRC Tool</p> <p>1.2 Design Rules</p> <p>1.3 Verification Report</p> <p>1.4 Rule Violations</p> <p>1.5 Error Markers</p>                                        |
| 2. Signoff verification flow          | <p>2.1 Signoff Flow</p> <p>2.2 Verification Process</p> <p>2.3 EDA Tool</p> <p>2.4 Project Requirements</p> <p>2.5 Verification Checklist</p>                         |
| 3. Signoff criteria                   | <p>3.1 Design Specifications</p> <p>3.2 Signoff Criteria</p> <p>3.3 Timing Constraints</p> <p>3.4 Physical Constraints</p> <p>3.5 Verification Standards</p>          |
| 4. Power consumption and distribution | <p>4.1 Dynamic Power</p> <p>4.2 Leakage Power</p> <p>4.3 Total Power</p> <p>4.4 Power Distribution Network (PDN)</p> <p>4.5 Power Density</p> <p>4.6 Power Budget</p> |
| 5. Power optimization opportunities   | <p>5.1 Clock Gating</p> <p>5.2 Multi-Vt Optimization</p> <p>5.3 Cell Sizing</p> <p>5.4 Power Gating</p>                                                               |

|                              |                                                                                                                                                 |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | 5.5 Buffer Optimization<br>5.6 Logic Optimization                                                                                               |
| 6. Signal integrity analysis | 6.1 Crosstalk Analysis<br>6.2 Noise Analysis<br>6.3 Signal Integrity Report<br>6.4 Delay Analysis<br>6.5 Glitch Analysis<br>6.6 Switching Noise |
| 7. Corrective actions        | 7.1 Buffer Insertion<br>7.2 Shielding<br>7.3 Cell Resizing<br>7.4 Routing Optimization<br>7.5 Power Grid Optimization<br>7.6 ECO Fixes          |
| 8. Signoff reports           | 8.1 STA Report<br>8.2 DRC Report<br>8.3 LVS Report<br>8.4 Power Report<br>8.5 Signal Integrity Report<br>8.6 EM/IR Report                       |
| 9. Signoff package           | 9.1 GDSII File<br>9.2 Netlist<br>9.3 SPEF File<br>9.4 SDC File<br>9.5 Signoff Reports<br>9.6 Tape-Out Documentation                             |

## Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning knowledge | 1.1 Principles of physical verification, including Design Rule Checking (DRC) and Layout Versus Schematic (LVS) verification.<br>1.2 Sign-off verification flow, criteria, and tape-out readiness requirements in ASIC physical design.<br>1.3 Power analysis concepts and power integrity.<br>1.4 Signal integrity fundamentals and reliability verification.<br>1.5 Methods for identifying, analyzing, and resolving DRC, LVS, timing, power, and reliability violations.<br>1.6 Interpretation of physical verification, timing, power, and reliability reports using EDA sign-off tools.<br>1.7 Procedures for reviewing, validating, and documenting final sign-off results and verification reports. |
| 2. Underpinning skills    | 2.1 Performing DRC and LVS verification to ensure layout                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>accuracy and design compliance</p> <p>2.2 Executing signoff verification procedures according to project requirements</p> <p>2.3 Analyzing power consumption, IR drop, leakage, and power integrity reports</p> <p>2.4 Performing signal integrity and reliability analysis.</p> <p>2.5 Applying corrective actions to improve power, timing, and design reliability</p> <p>2.6 Validating final verification results to ensure tape-out readiness</p> <p>2.7 Preparing signoff documentation and verification packages for final project delivery</p> |
| 3. Underpinning attitudes | <p>3.1 Commitment to occupational health and safety</p> <p>3.2 Promptness in carrying out activities</p> <p>3.3 Sincerity and honesty to duties</p> <p>3.4 Environmental concerns</p> <p>3.5 Eagerness to learn</p> <p>3.6 Tidiness and timeliness</p> <p>3.7 Respect for rights of peers and seniors in the workplace</p> <p>3.8 Good relationships with peers, subordinates and seniors in the workplace</p>                                                                                                                                            |
| 4. Resource implications  | <p>The following resources must be provided:</p> <p>4.1 Workplace (simulated or actual)</p> <p>4.2 Learning manual</p> <p>4.3 Specifications and work instructions</p>                                                                                                                                                                                                                                                                                                                                                                                    |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                          |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical aspects of competency | <p>Assessment must evidence that the candidate:</p> <p>1.1 performed DRC and LVS verification.</p> <p>1.2 performed signoff checks.</p> <p>1.3 analyzed power components.</p> <p>1.4 analyzed signal integrity and reliability.</p> <p>1.5 reviewed signoff results.</p> |
| 2. Methods of assessment          | <p>Methods of assessment should include:</p> <p>2.1 Written test</p> <p>2.2 Practical demonstration</p> <p>2.3 Oral questioning</p> <p>2.4 Portfolio (optional)</p>                                                                                                      |
| 3. Context of assessment          | <p>3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training.</p> <p>3.2 Assessment should be done by a nationally certified</p>                                                 |

|  |                                                  |
|--|--------------------------------------------------|
|  | assessor or occupation-specific industry expert. |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                     |                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------|
| <b>Unit of Competency:</b><br><b>EXECUTE ADVANCED DESIGN AND VERIFICATION TECHNIQUES</b>                                                                                                                                                                                                                                                                                                                                                                  | <b>Nominal Duration:</b><br>24 hrs. | <b>Unit Code:</b><br>SICIP-SCD-APDP-05-O |
| <b>Unit Descriptor:</b><br><br>This unit of competency covers the knowledge, skills and attitudes required to execute advanced design and verification techniques. It specifically includes the tasks of applying low-power design techniques, applying formal DFT and ATPG concepts, applying formal verification and assertion basics, performing LEC and formal debug, applying advanced flow optimization concepts and interpreting HLS fundamentals. |                                     |                                          |

### Elements and Performance Criteria

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                             | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Apply low-power design techniques.              | 1.1 <b><u>Low-power design requirements</u></b> are identified from the design specification.<br>1.2 Power intent is applied using appropriate <b><u>low-power design methodology</u></b> .<br>1.3 Clock gating, power gating, and multi-voltage techniques are implemented where required.<br>1.4 Low-power design reports are analyzed to verify power optimization.<br>1.5 Low-power implementation is validated against project requirements. |
| 2. Apply DFT and ATPG concepts.                    | 2.1 <b><u>DFT requirements</u></b> are identified according to design specifications.<br>2.2 Scan chains and test structures are incorporated into the design.<br>2.3 <b><u>ATPG tools and patterns</u></b> are generated using the appropriate EDA tool.<br>2.4 Test coverage and fault reports are analyzed to evaluate design quality.<br>2.5 DFT implementation results are verified before signoff.                                          |
| 3. Apply formal verification and assertion basics. | 3.1 <b><u>Formal verification objectives and design properties</u></b> are identified.<br>3.2 Assertions are developed according to functional requirements.<br>3.3 <b><u>Formal verification tools</u></b> are used to verify design behavior.<br>3.4 Verification failures are analyzed and corrected.<br>3.5 Formal verification results are reviewed and documented.                                                                          |
| 4. Perform LEC and formal debug.                   | 4.1 Logical Equivalence Checking (LEC) is performed between RTL and synthesized netlist.<br>4.2 LEC reports are analyzed to identify equivalence                                                                                                                                                                                                                                                                                                  |

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               | <p>mismatches.</p> <p>4.3 <b><u>Non-equivalent logic and verification issues</u></b> are debugged systematically.</p> <p>4.4 Design corrections are applied and revalidated using LEC.</p> <p>4.5 Final LEC results are reviewed before design signoff.</p>                                                                                                                                                                               |
| 5. Apply advanced flow optimization concepts. | <p>5.1 <b><u>Design flow bottlenecks</u></b> and optimization opportunities are identified.</p> <p>5.2 Advanced optimization techniques are applied to improve timing, area, and power.</p> <p>5.3 Design reports are analyzed to evaluate optimization results.</p> <p>5.4 <b><u>Flow parameters</u></b> are refined to improve implementation quality.</p> <p>5.5 Optimized design results are verified against project objectives.</p> |
| 6. Interpret HLS fundamentals.                | <p>6.1 High-Level Synthesis (HLS) concepts and design flow are interpreted.</p> <p>6.2 <b><u>C/C++-based design descriptions</u></b> are related to RTL implementation.</p> <p>6.3 HLS optimization directives and constraints are identified.</p> <p>6.4 HLS-generated RTL outputs and reports are interpreted.</p> <p>6.5 HLS applications and limitations are explained in the ASIC design flow.</p>                                   |

### Range of Variables

| Variable                         | Range<br>May include but not limited to:                                                                                                                                                  |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Low-power design requirements | <p>1.1 Power Budget</p> <p>1.2 Power Targets</p> <p>1.3 Performance Requirements</p> <p>1.4 Area Constraints</p> <p>1.5 Battery Life Requirements</p> <p>1.6 Low-Power Specifications</p> |
| 2. Low-power design methodology  | <p>2.1 UPF (Unified Power Format)</p> <p>2.2 CPF (Common Power Format)</p> <p>2.3 Power Domains</p> <p>2.4 Power States</p> <p>2.5 Isolation Strategy</p> <p>2.6 Retention Strategy</p>   |
| 3. DFT requirements              | <p>3.1 Scan Test</p> <p>3.2 Boundary Scan</p> <p>3.3 Built-In Self-Test (BIST)</p> <p>3.4 ATPG Requirements</p>                                                                           |

|                                                         |                                                                                                                                                                  |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                         | 3.5 Fault Coverage Targets<br>3.6 Test Specifications                                                                                                            |
| 4. ATPG tools and patterns                              | 4.1 Synopsys TestMAX<br>4.2 Siemens Tessent<br>4.3 Cadence Modus<br>4.4 Stuck-at Patterns<br>4.5 Transition Fault Patterns<br>4.6 Scan Patterns                  |
| 5. Formal verification objectives and design properties | 5.1 Functional Equivalence<br>5.2 Design Properties<br>5.3 Safety Properties<br>5.4 Liveness Properties<br>5.5 Design Constraints<br>5.6 Verification Objectives |
| 6. Formal verification tools                            | 6.1 Synopsys VC Formal<br>6.2 Cadence JasperGold<br>6.3 Siemens Questa Formal<br>6.4 OneSpin 360 DV<br>6.5 Formality<br>6.6 Verix                                |
| 7. Non-equivalent logic and verification issues         | 7.1 Logic Mismatch<br>7.2 Missing Logic<br>7.3 Signal Mapping Errors<br>7.4 Constraint Issues<br>7.5 RTL Coding Errors<br>7.6 Netlist Errors                     |
| 8. Design flow bottlenecks                              | 8.1 Timing Bottlenecks<br>8.2 Area Bottlenecks<br>8.3 Power Bottlenecks                                                                                          |
| 9. Flow parameters                                      | 9.1 Timing Constraints<br>9.2 Placement Parameters<br>9.3 CTS Parameters<br>9.4 Routing Parameters<br>9.5 Optimization Settings<br>9.6 Power Constraints         |
| 10. C/C++-based design descriptions                     | 10.1 C Language<br>10.2 C++ Language<br>10.3 Functions<br>10.4 Loops<br>10.5 Arrays<br>10.6 Data Types                                                           |

## Curricular Content Guide

Advanced Physical Design Program: RTL to GDSII



|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning knowledge | <p>1.1 Fundamentals of low-power VLSI design and power optimization methodologies.</p> <p>1.2 Design for Testability (DFT) concepts, scan chain architecture.</p> <p>1.3 Principles of formal verification, assertion-based verification (ABV), and property checking techniques.</p> <p>1.4 Logical Equivalence Checking (LEC) concepts, RTL-to-netlist equivalence verification.</p> <p>1.5 Advanced optimization techniques for improving timing, area, power.</p> <p>1.6 Interpretation of EDA reports related to power, DFT, formal verification, LEC, and design optimization.</p> <p>1.7 Fundamentals of High-Level Synthesis (HLS).</p>                                                    |
| 2. Underpinning skills    | <p>2.1 Applying low-power design techniques to optimize power consumption and efficiency</p> <p>2.2 Implementing DFT and ATPG concepts to improve design testability and fault coverage</p> <p>2.3 Applying formal verification and assertion techniques to validate functional correctness</p> <p>2.4 Performing Logical Equivalence Checking (LEC) and systematic formal debugging</p> <p>2.5 Applying advanced flow optimization techniques for timing, area, and power improvement</p> <p>2.6 Understanding HLS optimization directives, generated RTL outputs, and design reports</p> <p>2.7 Validating advanced design implementations against project requirements and signoff criteria</p> |
| 3. Underpinning attitudes | <p>3.1 Commitment to occupational health and safety</p> <p>3.2 Promptness in carrying out activities</p> <p>3.3 Sincerity and honesty to duties</p> <p>3.4 Environmental concerns</p> <p>3.5 Eagerness to learn</p> <p>3.6 Tidiness and timeliness</p> <p>3.7 Respect for rights of peers and seniors in the workplace</p> <p>3.8 Good relationships with peers, subordinates and seniors in the workplace</p>                                                                                                                                                                                                                                                                                     |
| 4. Resource implications  | <p>The following resources must be provided:</p> <p>4.1 Workplace (simulated or actual)</p> <p>4.2 Learning materials</p> <p>4.3 Specifications and work instructions</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical aspects of competency | <p>Assessment must evidence that the candidate:</p> <ul style="list-style-type: none"> <li>1.1 applied low-power design techniques.</li> <li>1.2 applied DFT and ATPG concepts.</li> <li>1.3 applied formal verification and assertion basics.</li> <li>1.4 performed LEC and formal debug.</li> <li>1.5 applied advanced flow optimization concepts.</li> <li>1.6 interpreted HLS fundamentals.</li> </ul> |
| 2. Methods of assessment          | <p>Methods of assessment should include:</p> <ul style="list-style-type: none"> <li>2.1 Written test</li> <li>2.2 Practical demonstration</li> <li>2.3 Oral questioning</li> <li>2.4 Portfolio (optional)</li> </ul>                                                                                                                                                                                        |
| 3. Context of assessment          | <ul style="list-style-type: none"> <li>3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</li> <li>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</li> </ul>                                                                                       |

|                                                                                                                                                                                                                                                                                                                                                                                                                          |                                     |                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------|
| <b>Unit of Competency:</b><br><b>PERFORM RTL-TO-GDSII PHYSICAL DESIGN PROJECT</b>                                                                                                                                                                                                                                                                                                                                        | <b>Nominal Duration:</b><br>32 hrs. | <b>Unit Code:</b><br>SICIP-SCD-APDP-06-O |
| <b>Unit Descriptor:</b><br><br>This unit of competency covers the knowledge, skills and attitudes required to perform RTL-to-GDSII physical design project. It specifically includes the tasks of planning RTL-to-GDSII implementation, executing the complete physical design flow, validating design quality and signoff results, preparing project documentation and deliverables, presenting and defend design work. |                                     |                                          |

#### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                          | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Plan RTL-to-GDSII implementation.            | 1.1 Project requirements and design specifications are reviewed before implementation.<br>1.2 Design flow, milestones, and implementation tasks are planned according to project objectives.<br>1.3 Required <b><u>EDA tools, technology libraries, and design resources</u></b> are prepared.<br>1.4 Project constraints, timelines, and deliverables are identified and documented.<br>1.5 <b><u>Project plan</u></b> is reviewed and approved before implementation.   |
| 2. Execute the complete physical design flow.   | 2.1 RTL design is synthesized according to project requirements.<br>2.2 Floor planning, placement, clock tree synthesis, routing, and physical optimization are performed.<br>2.3 Physical verification and timing analysis are conducted throughout the implementation process.<br>2.4 <b><u>Design issues</u></b> are identified and resolved during implementation.<br>2.5 <b><u>Final GDSII database</u></b> is generated following successful design implementation. |
| 3. Validate design quality and signoff results. | 3.1 Timing, power, area, and physical verification reports are reviewed.<br>3.2 DRC, LVS, STA, and <b><u>signoff results</u></b> are validated against project requirements.<br>3.3 <b><u>Design quality metrics</u></b> are evaluated to ensure implementation objectives are achieved.<br>3.4 Outstanding design issues are corrected before final approval.<br>3.5 Final design is verified for tape-out readiness.                                                    |
| 4. Prepare project documentation and            | 4.1 Project reports and technical documentation are prepared using standard formats.                                                                                                                                                                                                                                                                                                                                                                                      |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| deliverables.                      | <p>4.2 Design files, scripts, reports, and signoff results are compiled for submission.</p> <p>4.3 Project activities, implementation results, and design decisions are documented.</p> <p>4.4 Final deliverables are reviewed for completeness and accuracy.</p> <p>4.5 <b><u>Project documents</u></b> are submitted according to organizational requirements.</p>                                                                                                        |
| 5. Present and defend design work. | <p>5.1 Project objectives, design methodology, and implementation flow are presented clearly.</p> <p>5.2 Design decisions and implementation results are explained using technical evidence.</p> <p>5.3 <b><u>Questions and technical feedback</u></b> are addressed professionally.</p> <p>5.4 Design improvements and lessons learned are discussed during the presentation.</p> <p>5.5 Final project presentation is completed according to assessment requirements.</p> |

### Range of Variables

| Variable                                                 | Range                                                                                                                                                                                |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          | May include but not limited to:                                                                                                                                                      |
| 1. EDA tools, technology libraries, and design resources | <p>1.1 Cadence Innovus</p> <p>1.2 Cadence Genus</p> <p>1.3 Synopsys Design Compiler</p> <p>1.4 Standard Cell Libraries</p> <p>1.5 Technology Files</p> <p>1.6 Design Workstation</p> |
| 2. Project plan                                          | <p>2.1 Project Schedule</p> <p>2.2 Resource Plan</p> <p>2.3 Task List</p> <p>2.4 Implementation Plan</p> <p>2.5 Review Checklist</p> <p>2.6 Approval Record</p>                      |
| 3. Design issues                                         | <p>3.1 Timing Violations</p> <p>3.2 DRC Violations</p> <p>3.3 LVS Errors</p> <p>3.4 Congestion Issues</p> <p>3.5 Power Integrity Issues</p> <p>3.6 Routing Violations</p>            |
| 4. Final GDSII database                                  | <p>4.1 GDSII File</p> <p>4.2 Tape-Out Database</p> <p>4.3 Sign-off Reports</p> <p>4.4 Final Netlist</p>                                                                              |

|                                     |                                                                                                                                                                               |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | 4.5 SPEF File<br>4.6 Design Database                                                                                                                                          |
| 5. Signoff results                  | 5.1 DRC Results<br>5.2 LVS Results<br>5.3 STA Results<br>5.4 Signal Integrity Results<br>5.5 EM/IR Analysis Results<br>5.6 Sign-off Checklist                                 |
| 6. Design quality metrics           | 6.1 Timing Closure<br>6.2 Area Utilization<br>6.3 Power Consumption<br>6.4 Worst Negative Slack (WNS)<br>6.5 Total Negative Slack (TNS)<br>6.6 QoR (Quality of Results)       |
| 7. Project documents                | 7.1 Submission Package<br>7.2 Approval Documents<br>7.3 Delivery Checklist<br>7.4 Project Archive<br>7.5 Release Notes<br>7.6 Handover Documents                              |
| 8. Questions and technical feedback | 8.1 Technical Questions<br>8.2 Design Review Comments<br>8.3 Clarification Requests<br>8.4 Improvement Suggestions<br>8.5 Assessment Feedback<br>8.6 Reviewer Recommendations |

## Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning knowledge | 1.1 Identify RTL-to-GDSII ASIC design flow, project lifecycle, and implementation methodology.<br>1.2 Project planning principles and deliverables.<br>1.3 Use of EDA tools, technology libraries, design constraints.<br>1.4 Physical design implementation concepts and physical verification.<br>1.5 Design quality metrics, sign-off criteria.<br>1.6 Technical documentation standards for design reports.<br>1.7 Principles of technical presentation, design review, and evidence-based communication. |
| 2. Underpinning skills    | 2.1 Planning RTL-to-GDSII implementation activities based on project requirements and design specifications<br>2.2 Preparing EDA tools, technology libraries, and design                                                                                                                                                                                                                                                                                                                                      |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <p>resources for implementation</p> <p>2.3 Executing the complete ASIC physical design flow from synthesis to GDSII generation</p> <p>2.4 Applying physical verification, timing analysis, and design optimization techniques</p> <p>2.5 Validating timing, power, area, DRC, LVS, STA, and signoff results against project requirements</p> <p>2.6 Preparing technical documentation, design reports, scripts, and project deliverables</p> <p>2.7 Presenting and defending design methodology, implementation results, and technical decisions.</p> |
| 3. Underpinning attitudes | <p>3.1 Commitment to occupational health and safety</p> <p>3.2 Promptness in carrying out activities</p> <p>3.3 Sincerity and honesty to duties</p> <p>3.4 Environmental concerns</p> <p>3.5 Eagerness to learn</p> <p>3.6 Tidiness and timeliness</p> <p>3.7 Respect for rights of peers and seniors in the workplace</p> <p>3.8 Good relationships with peers, subordinates and seniors in the workplace</p>                                                                                                                                        |
| 4. Resource implications  | <p>The following resources must be provided:</p> <p>4.1 Workplace (simulated or actual)</p> <p>4.2 Learning materials</p> <p>4.3 Specifications and work instructions</p>                                                                                                                                                                                                                                                                                                                                                                             |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical aspects of competency | <p>Assessment must evidence that the candidate:</p> <p>1.1 planned RTL-to-GDSII implementation.</p> <p>1.2 executed the complete physical design flow.</p> <p>1.3 validated design quality and signoff results.</p> <p>1.4 prepared project documentation and deliverables.</p> <p>1.5 presented and defended design work.</p> |
| 2. Methods of assessment          | <p>Methods of assessment should include:</p> <p>2.1 Written test</p> <p>2.2 Practical demonstration</p> <p>2.3 Oral questioning</p> <p>2.4 Portfolio (optional)</p>                                                                                                                                                            |
| 3. Context of assessment          | <p>3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</p> <p>3.2 Assessment should be done by a nationally certified</p>                                                                                                        |

|  |                                                  |
|--|--------------------------------------------------|
|  | assessor or occupation-specific industry expert. |
|--|--------------------------------------------------|

**END OF COMPETENCY STANDARD**

## **Workshop/Lab Facility Standard**

|                            |                                                |
|----------------------------|------------------------------------------------|
| <b>Course Name:</b>        | Advanced Physical Design Program: RTL to GDSII |
| <b>Number of Trainees:</b> | 25                                             |

### **Course-wise Training Space:**

- Computer Lab : 400 sft
- Workshop : 700-800 sft

### **Major Training Equipment and Training Facilities:**

| <b>S.N.</b> | <b>Major Training Equipment and facilities</b> | <b>Required facilities</b> |
|-------------|------------------------------------------------|----------------------------|
| 1.          | Computer/Laptops                               | 01                         |
| 2.          | Printer                                        | 01                         |
| 3.          | Multimedia projector                           | 01                         |
| 4.          | High-performance Desktop/Laptop                | 25                         |
| 5.          | EDA Software Licenses (Floating/User Licenses) | 25                         |
| 6.          | Linux Operating System                         | 25                         |
| 7.          | Central EDA License Server                     | 01                         |
| 8.          | Central File/Storage Server (NAS/Server)       | 01                         |
| 9.          | Gigabit Network Switch (48-Port)               | 01                         |
| 10.         | Enterprise Router                              | 01                         |
| 11.         | RTL Simulation Environment                     | 02                         |
| 12.         | Static Timing Analysis (STA) Tool              | 25                         |
| 13.         | Physical Design Tool                           | 25                         |
| 14.         | Physical Verification (DRC/LVS) Tool           | 25                         |
| 15.         | Power & Signal Integrity Analysis Tool         | 25                         |
| 16.         | Formal Verification Tool                       | 25                         |
| 17.         | Version Control Server (Git)                   | 01                         |
| 18.         | Laser Pointer                                  | 01                         |



The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Advanced Physical Design Program: RTL to GDSII, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.